



Environmental Illness

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EPIDEMIOLOGY

Emergency department visits in the United States for exertional heat illness (EHI) increased 133% during the period 1997–2006.¹ Cases of EHI consisted of heat exhaustion (72.7%), heat syncope (9.7%), heat cramps (5.4%), and heat stroke (1.8%). Patients younger than 19 years (47.6%) and males (71.9%) accounted for a disproportionate percentage of EHI cases. The highest incidence of exercise-related heat illness in the United States occurs in August.^{2,3,4} From 1990 to 2010, there were 49 heat-related deaths in high school and college football, more than any other sport.⁵ Of these fatalities, 86% were lineman.

DEFINITION

EHI represents a spectrum varying from mild symptoms to medical emergencies. Environmental heat stress is best estimated by using the wet bulb globe temperature (WBGT), which takes into account radiant heat from the sun, ambient temperature, and relative humidity.⁶ However, heat stress is more commonly reported as the more intuitively understood heat index (see <http://www.nws.noaa.gov/om/heat/index.shtml>).

Exercising muscle is relatively inefficient, converting ATP into muscular work with less than 25% efficiency, resulting in marked heat production, the amount of which increases with exercise intensity.⁷ The initial responses to increasing core temperature include increased sweat, cardiac output, vasodilation, and redistribution of blood to the muscles and skin.⁸ Heat dissipation during exercise is accomplished primarily by evaporation of sweat off the skin. High humidity will impede sweat evaporation. Dehydration (loss of 1% to 2% of body weight) is sufficient to impair effective sweat production.⁹

Forms of EHI can be subdivided by the presence of elevated body temperature. Elevated temperature EHI includes heat exhaustion and heat stroke. Normal body temperature EHI includes exercise-associated muscle cramps, heat syncope, exertional rhabdomyolysis, exertional sickle crisis, and exercise-associated hyponatremia.

EXERTIONAL HEAT EXHAUSTION

Pathophysiology

Exertional heat exhaustion (EHE) is likely due to a combination of dehydration, central nervous system fatigue, depletion of muscle

energy stores, and in some cases, electrolyte imbalance.^{7,10} Heat exhaustion typically occurs in high-temperature environments but may also occur in moderate- to low-temperature environments with high humidity.

Diagnosis

The diagnosis of EHE is made clinically. Symptoms include fatigue, dizziness, muscle cramps, headache, and nausea. Mild mental status changes may be observed (mild confusion, irritability, and emotional lability). Common physical signs include pallor and weakness. Vital signs will show elevated heart rate and body temperature ($<39.5^{\circ}\text{C}$) with normal to low blood pressure.

Laboratory Findings

Laboratory testing is of little value except in cases of suspected hyponatremia (described later). Elevated urine-specific gravity and hemoconcentration reflect dehydration.

Differential Diagnosis

Other conditions that may resemble EHE include concussion, hypothermia, viral illness, alcohol or other drug use, and cardiac problems.

Treatment

The athlete should rest in a cool place. Oral rehydration is the initial treatment of choice, with the goal of replacing 1.5 L of fluid for every kilogram of body weight lost.¹¹ If body weight is unknown, adequate rehydration is evidenced by improved vital signs, ability to urinate, less orthostasis, and generally “feeling better.” Cooled fluids containing less than 8% carbohydrate are absorbed more rapidly than plain water.^{12–15} Rehydration fluids containing sodium appear to restore total body water levels more completely than water alone.^{16,17} Intravenous (IV) hydration may be required for vomiting or significant orthostatic issues. Recommended IV solutions are normal saline (NS) and 5% dextrose in NS (D5NS).¹⁸ The authors prefer D5NS to help limit the depletion of glycogen stores in the post-exercise hypermetabolic state. The infusion of IV fluids in heat exhaustion patients does not improve dehydration quicker than oral fluid, and IV infusions are prohibited by the World Anti-Doping Agency for athletes who are able to drink.^{19,20}

Return to Play

No evidence-based guidelines have been developed for return to play (RTP) after heat exhaustion. Our recommendations are

Abstract

Environmental illnesses are encountered by athletes participating in intense or prolonged activity in conditions of high heat and humidity, cold, or at high altitude. Permanent disability and death are possible with the more severe forms of environmental illness. In some cases, intrinsic characteristics, improper training methods, or the use of certain drugs may predispose athletes to environmental illness. Illnesses due to environmental exposures are largely preventable.

Keywords

exertional heat illness
heat stroke
hyponatremia
sickle trait
hypothermia
frostbite
high-altitude illness

BOX 21.1 Return-to-Play Criteria After Heat Exhaustion

- Minimum of 24–48 h of rest before returning to activity
- No headache, gastrointestinal symptoms, or muscle soreness
- Normal tolerance of orally ingested food and fluids
- Normal serum electrolytes (if the athlete was known to have an electrolyte disturbance)
- Normal-appearing urine

listed in [Box 21.1](#). RTP should address risk factors including deconditioning, chronic dehydration, habitual fatigue, and excess body weight.

Prevention

Several sports organizations recommend modifying outdoor physical activity when heat index values are in the “extreme caution” or “danger” zones.^{21–23} Modifications include reducing the number of practices, scheduling practices during cooler times of the day, shedding equipment, and increasing access to water. These recommendations pertain especially to North American football, which is responsible for more cases of heat-related illness and deaths than any other sport.^{24–26} In addition, athletes should be educated about gauging hydration status based on urine color; urine the color of apple juice reflects dehydration, whereas lemonade-colored urine reflects adequate hydration.

Acclimatization to hot environments occurs by a number of physiologic adaptations, including initiation of sweating at a lower core temperature and thirst at a lower serum osmolality, increased sweat rate, increased sodium absorption from sweat and urine, and expanded plasma volume.^{7,18,27} The time required for acclimatization depends on age, baseline level of conditioning, practice and equipment requirements, and environmental factors. College athletes have been shown to acclimatize in as few as 12 days of practice.²⁸ Younger athletes typically require a longer period of acclimatization.^{29,30}

EXERTIONAL HEAT STROKE

Exertional heat stroke (EHS) occurs when the body is not able to compensate for environmental heat stress combined with body heat produced by exercise. EHS is defined as an elevated core temperature (usually $>40^{\circ}\text{C}$ [104°F]) associated with signs of organ system failure. Central nervous system changes (altered mental status) are often the first markers of EHS.³¹ EHS usually occurs in otherwise healthy, fit persons participating in high-intensity exercise in the heat.

Once homeostatic mechanisms fail to maintain a stable core temperature, continued heat stress will increase the shunting of blood flow from the core to the periphery, leading to organ hypoxia and acidosis. Resulting anaerobic metabolism depletes carbohydrate stores, which compounds organ damage.^{8,32} Organ damage, particularly to the gastrointestinal tract, allows endotoxins to enter the circulation, resulting in hypotension, and the release of inflammatory cytokines results in decreased cerebral blood flow and neuronal injury.⁸ Muscle injury results in increased serum [Ca] and [K]. Death usually occurs as a result of shock.

BOX 21.2 Return-to-Play Criteria After Heat Stroke

- Demonstration of normal mental status
- Return to normal dietary and elimination habits
- Normal variation in daily body temperature
- Normal complete blood cell count, blood urea nitrogen/creatinine, anion gap, liver enzymes, creatine phosphokinase, and urinalysis, including specific gravity
- Normal echocardiogram
- Persons with complicated heat stroke (i.e., significant organ failure and extensive rhabdomyolysis) should refrain from exercise for 1 month
- Persons with uncomplicated heat stroke (i.e., a brief emergency department or hospital stay) should refrain from exercise for 1 week
- Begin exercise in a cool environment, gradually increasing the duration, intensity, and heat exposure
- Athletes who demonstrate heat tolerance after 2–4 weeks of training may return to competition

Diagnosis

The person with EHS may start out as dizzy, nauseated, and confused, but lethargy, obtundation, or unresponsiveness indicate more advanced, critical EHS. Physical examination typically reveals hot, sweaty skin, rapid pulse, and low blood pressure. Body temperature is above 40°C (104°F). Rectal temperature measurement is the most reliable method of monitoring core body temperature in the exercising athlete.²⁶

Treatment

The keys to a favorable outcome for a person with EHS are (1) early recognition and (2) a minimal delay in initiating cooling.³³ Cooling should never wait for rectal temperature measurement in an athlete suspected of having EHS. The first responder should initiate basic cardiopulmonary resuscitation if necessary and activate EMS. Clothing and equipment should be removed immediately. Immersion should occur first if removing cumbersome equipment would delay treatment.³⁴ Immersion of the entire body in cold water is the treatment of choice.³⁵ Alternatives include pouring cold ice water continuously over the entire body, placing ice bags on the scalp, neck, axillae, groin, and popliteal fossae, and performing ice massage of large muscles.³⁶ Cooling continues until the rectal temperature reaches 38.6°C (101.4°F) to prevent overcooling.³⁷

Return to Play

Recommendations for RTP criteria are summarized in [Box 21.2](#).^{38,39} No validated tests are available to measure responses to graded exercise in the heat to assist in making return-to-play decisions.⁴⁰

EXERCISE-ASSOCIATED MUSCLE CRAMPS

Pathophysiology

Exercise-associated muscle cramps (EAMC), commonly known as *heat cramps*, are involuntary, painful muscular contractions during exercise. Several hypotheses have been proposed to explain EAMC, including excessive sodium losses, muscle fatigue, and

central nervous system fatigue.⁴¹⁻⁴³ Prospective studies have failed to correlate sodium losses with the onset of EAMC.⁴¹ Fatigue causes an increased firing rate of type Ia and II (excitatory) muscle spindle afferents and a decrease in afferent activity from the Golgi tendon organ (inhibitory) that results in sustained alpha motor neuron firing.⁴³ Poor conditioning may be result in premature fatigue.

Diagnosis

EAMC are usually self-evident. The athlete who has been exercising vigorously experiences painful cramping and is unable to function normally.

Laboratory Findings

In the majority of cases, electrolytes, pH, and muscle creatine phosphokinase (CPK) levels are within normal limits.

Differential Diagnosis

It is important to exclude drug effects as a cause of muscle tetany, including the use of lipid lowering and antihypertensive drugs, beta-agonists, oral contraceptives, insulin, and creatine.⁴⁴

Treatment

The most effective treatment for EAMC appears to be passive stretching, which presumably increases the inhibitory activity of the Golgi tendon organ.^{41,43} Cooling the muscles with ice massage may provide additional relief. Little evidence indicates that ingestion of an electrolyte solution or pickle juice consistently resolves EAMC.

Prevention

Athletes who are prone to cramping should be encouraged to consume more than normal amounts of salt in their diet and to use oral hydration fluids that contain sodium during and after practice. Adequate conditioning and improved rest may also help prevent EAMC.^{43,44}

Return to Play

Athletes may return to the field of play after the resolution of muscle cramps if they are able to perform sports-specific drills without recurrence of EAMC.

HEAT SYNCOPE/EXERCISE ASSOCIATED COLLAPSE

Pathophysiology

Heat syncope (HS) and exercise-associated collapse (EAC) occur immediately following the stress of heat exposure and/or exercise. Upon cessation of exercise (i.e., the finish line of a marathon), the absence of rhythmic muscular contraction combined with dehydration limits, venous return, and thus cerebral perfusion pressure, results in syncope in susceptible individuals.⁴⁵ Syncope may occur similarly after prolonged standing in the heat or sudden rising from a sitting or lying position after exercise.

Diagnosis

Syncope that occurs upon stopping prolonged exercise in the absence of any other cause of syncope is EAC. Likewise, syncope

in the heat after prolonged standing or sudden postural changes is likely HS. Orthostatic blood pressure changes may be present.

Laboratory Findings

Laboratory findings should be normal.

Differential Diagnosis

When syncope occurs during or immediately after exercise, a cardiac etiology must be assumed until proven otherwise. Seizure with postictal confusion can look very similar to HS and EAC. EHS must also be considered in EAC but in HS/EAC, the core temperature will be normal and individuals will respond to positional support. Hypoglycemia associated with diabetes must also be considered in the syncopal athlete not responding to supportive care.

Treatment

Supportive care to restore normal perfusion of the brain is essential. Placing patients in the Trendelenburg position for 15 to 30 minutes will help to restore cerebral blood flow. Squeezing crossed legs together and making a strong fist have been useful in increasing venous return and reestablishing adequate perfusion.⁴⁶

Prevention

Adequate hydration with sufficient salt intake is important to maintaining normal fluid volume. Continuing to walk after completing a running race can minimize the risk for EAC. Moving or contracting the muscles of the legs during prolonged standing in the heat can help to maintain adequate perfusion.

Return to Play

Athletes may RTP after resolution of symptoms as long as they have reestablished normal blood pressure without orthostasis and have no signs of dehydration.

EXERTIONAL RHABDOMYOLYSIS

Pathophysiology

Exertional rhabdomyolysis (ER) is characterized by the breakdown of skeletal muscle during and after exercise that exceeds the working ability of the muscles.⁴⁷

Excessive exercise will result in elevated intracellular free ionized calcium (Ca) especially in the mitochondria, compounded by depletion of ATP causing dysfunction in the Ca-ATPase pump.⁴⁸ This elevated calcium results in the activation of proteases and oxygen free radicals that cause myocyte death. Muscle injury causes intense muscular pain, weakness, swelling and elevated levels of serum CPK at least five times the upper limit of normal.⁴⁹ Risk factors for ER include dehydration, acidosis, excess ambient temperature, and hypoxia. In persons with ER, large amounts of myoglobin, proteases, and inflammatory mediators enter the circulation. Filtration of myoglobin results in glomerular injury, and proteases damage the respiratory epithelium, which may lead to respiratory distress. Rarely, inborn errors of metabolism may present with ER in apparently healthy persons.^{48,50}

Diagnosis

Affected persons typically present with severe muscle pain especially those muscles involved in the extreme exercise. The muscles are tender to palpation and may be swollen. Severely affected patients may present with tachypnea and edema. Severe consequences include renal failure, disseminated intravascular coagulopathy, and acidosis.

Laboratory Findings

CPK levels usually exceed 20,000 IU/L or five times the upper limit of normal. Myoglobinuria may portend renal injury, but has also been report in asymptomatic endurance athletes.⁵¹ Severe cases can have elevated serum potassium, positive D-dimer, elevated blood urea nitrogen (BUN), and creatinine.⁵²

Differential Diagnosis

ER must be distinguished from compartment syndrome and pyomyositis. *Exertional myositis* refers to moderately elevated CPK levels ($\leq 20,000$ IU/L) in an athlete experiencing mild-to-moderate muscle pain and weakness without evidence of renal or metabolic injury.^{51,53}

Treatment

Treatment usually requires hospitalization for the correction of acidosis, careful hydration to aid in the clearance of myoglobin without inducing fluid overload, monitoring renal function, and supporting cardiac and pulmonary function. If CPK does not improve after 72 hours, renal consultation should be considered. Persons who meet the following criteria may be treated as outpatients with close follow-up: (1) CPK less than 15,000 IU/L; (2) normal renal function *and* mild dehydration; and (3) the absence of sickle cell trait, infectious disease, or an underlying metabolic syndrome.⁵⁴ These persons should be encouraged to recover in a cool environment, refrain from exercise, and drink plenty of fluids.

Return to Play

Our RTP criteria are listed in [Box 21.3](#).^{38,54,55}

Prevention

ER can be prevented by gradually increasing training intensity and volume, and by ensuring adequate hydration and recovery between exercise bouts. ER has occurred in persons involved in activities, such as physical education, band, cheerleading, drill team, and Reserve Officers' Training Corps, especially when excessive exercise is used as a reprimand. Persons involved in these activities should be provided free access to water and sufficient rest during exercise, and they should be educated about symptoms and signs of heat illness and muscle injury.

EXERTIONAL SICKLING COLLAPSE

Pathophysiology

Sickle cell trait (SCT) is defined as the presence of 30% to 40% hemoglobin-S. Persons with SCT are normally asymptomatic. SCT occurs in 7% to 10% of African Americans and is common in persons of Mediterranean ancestry. Football players with SCT

BOX 21.3 Return-to-Play Criteria After Exertional Rhabdomyolysis

- Absence of muscle soreness or tenderness
- Normal strength
- Creatine phosphokinase less than five times the upper limit of normal
- Normal urinalysis, blood urea nitrogen/creatinine
- Persons with severe rhabdomyolysis (i.e., significant cardiac, pulmonary, and renal dysfunction) should not participate in any activity for 2–4 weeks after hospital discharge
- Persons with mild rhabdomyolysis may return to light activity when the aforementioned criteria are met
- Follow-up medical examination after 1 week
- May return to competition after tolerating 2–3 weeks of gradually increasing activity and demonstration of normal creatine phosphokinase values

Data from George M, Delgaudio A, Salhanick SD. Exertional rhabdomyolysis—when should we start worrying? Case reports and literature review. *Pediatr Emerg Care*. 2010;26:864–866; and O'Connor FG, Brennan FH, Campbell W, et al. Return to physical activity after exertional rhabdomyolysis. *Curr Sports Med Rep*. 2008;7(6):328–331.

TABLE 21.1 Exertional Sickling Collapse Versus Exercise-Associated Muscle Cramps

Feature	Exertional Sickling Collapse	Exercise-Associated Muscle Cramps
Muscles	Weakness > pain No spasm (tetany)	Cramping pain $\geq$ weakness, spasm
Behavior	Alert, may become listless, rapid breathing	Listless, irritable, may become combative
Temperature	Normal	Normal or elevated
Timing in exercise	Often occurs early in training session	Nearly always occurs late in training session

See references 57, 58.

have a 37 to 67 times higher risk of exertion-related death compared with non-SCT athletes.^{5,56}

Heat stress, dehydration, and intense exercise especially at high altitude can cause severe hypoxemia, acidosis, hyperthermia, and red cell dehydration, which may induce sickling of red blood cells, leading to ESC.^{57,58} Persons with SCT appear to be at greatest risk when they perform short bursts of repetitive, high-intensity activity (e.g., sprints), especially during the preseason when they be deconditioned.

Diagnosis

Athletes who experience ESC have sudden onset of extreme weakness and may collapse on the field, appearing anxious, and tachypneic. Features that help distinguish an athlete with ESC from an athlete with exercise-associated muscle cramps are outlined in [Table 21.1](#).⁵⁸

Laboratory Findings

The presence and percentage of sickle hemoglobin (HgbS) is determined by hemoglobin electrophoresis. In the acutely ill

athlete, a complete blood cell count reveals anemia, and a blood smear reveals classic sickled red blood cells. Acutely, arterial blood gas findings reveal a partially compensated metabolic acidosis (low carbon dioxide, high pH, and increased anion gap). Muscle CPK is elevated and may continue to rise, akin to elevations observed in persons with ER.

Differential Diagnosis

Conditions that may mimic ESC include collapse from cardiac abnormalities (conduction or structural), asthma, heat exhaustion, or EHS. Cardiac arrest or heat stroke are unconscious or obtunded, whereas an athlete who collapses with ESC is usually lucid.⁵⁸

Treatment

An athlete with suspected SCT-associated ER should not be allowed to perform any further exercise that day. EMS should be summoned and the athlete's condition should be treated as a "medical emergency." On-field measures should include oral rehydration, removal of the athlete to a cool environment, and the expeditious use of supplemental oxygen and IV fluids if available.

Return to Play

For general guidelines, medical staff may refer to the guidelines for RTP after ER (see Box 21.3).^{38,54,55}

Prevention

The sickle trait status should be known in all athletes born in states that mandate determination of hemoglobin status as part of the newborn metabolic screen. Universal screening for sports beyond the newborn period is controversial.⁵⁸ Currently, the NCAA mandates screening of all athletes entering collegiate programs.⁵⁶ In other settings, selective screening of high-risk groups, particularly African Americans, may be warranted. Athletes with SCT should be counseled to set their own pace, slowly build-up training with longer recovery between exercise bouts, maintain good hydration, avoid extreme performance tests, and to stop exercising immediately if they experience unusual muscle pain or weakness.⁵⁸ The authors feel these guidelines should apply to all athletes regardless of whether their sickle trait status is known.

EXERCISE-ASSOCIATED HYPONATREMIA

Pathophysiology

Exercise-associated hyponatremia (EAH) is defined as the serum sodium (Na) below normal reference values (usually 135 mmol/L) during or 24 hours following physical activity.⁵⁹ EAH has been most frequently reported in endurance athletes, but has also been reported in the soldiers, hikers, and football players.⁶⁰⁻⁶³ EAH has a spectrum of severity from mild cases complaining of nausea, dizziness, and "not feeling right" to vomiting, headaches, and confusion, and ultimately to coma, seizures, pulmonary edema, and death. The most common etiology of EAH is exercising while over hydrating, which may be related to lack of education about proper hydration or attempts to

treat EAMC.^{64,65} The syndrome of inappropriate antidiuretic hormone secretion (SIADH) will cause fluid retention when diuresis is needed. Tightly regulated osmotic control of body fluids can be overridden by nonosmotic stimuli, including nausea/vomiting, pain, exercise, elevated body temperature.⁶⁶⁻⁷⁰ Much less frequently, EAH can be related to excessive sweat sodium losses with relatively normal fluid replacement with hypotonic fluid.⁵⁹

Diagnosis

Increased body weight with decreased production of urine in an endurance athlete with the symptoms described above, especially with adequate water consumption, should raise suspicion of EAH.

Laboratory Findings

Clinical suspicion of EAH should prompt evaluation of a basic metabolic panel. Mild or asymptomatic cases usually have a serum Na between 130 to 135 mmol/L while more severe cases are ≤ 129 mmol/L. Associated elevation urine osmolality indicates SIADH.

Differential Diagnosis

Altered mental status associated with exercise may be caused alternatively by EHS, HS, or EAC, head trauma, seizure, drug ingestion, and diabetic ketoacidosis.

Treatment

In mild cases, fluid restriction with free access to salty snacks and broth will prevent the progression to more severe hyponatremia. Reestablishment of the body's homeostatic mechanisms is indicated with improving serum Na and the production of dilute urine. In severe cases with significantly altered mental status, an IV bolus of 100 mL 3% hypertonic saline should be given and can be repeated twice at 10-minute intervals for lack of clinical improvement.⁶⁷ Rapid correction of serum Na in EAH poses minimal risk for central pontine myelinolysis, which is seen with rapid correction of chronic hyponatremia.⁵⁹

Return to Play

The athlete with EAH may return to activity when asymptomatic, the serum Na is normal, and urine production is normal. Proper hydration practices must be instituted and exercise should be done at a moderate intensity and duration, at least during the initial 1 to 2 weeks, avoiding extreme environmental conditions.

Prevention

Proper hydration is the best prevention and intervention. While some advocate drinking to thirst, replacing sweat loss using an individualized objective strategy will help novice athletes especially who may have difficulty in distinguishing thirst from fatigue. The sweat rate should be measured under different environmental conditions (Box 21.4). Secondary prevention can be achieved by measuring body weight before and after an event. Those who fail to lose weight during a marathon are seven times more likely to develop exercise-associated hyponatremia.^{65,70} Medical and

BOX 21.4 Sweat Rate Measurement

1. Go to the bathroom and urinate/defecate if needed.
 2. Towel off any sweat if needed.
 3. Measure weight in minimal clothes/naked (PREWEIGHT).
 4. Get dressed.
 5. Perform planned exercise/practice for 1 hour with access to water in a container marked for volume.
 6. Towel off any sweat.
 7. Measure weight in minimal clothes/naked (POSTWEIGHT).
 8. Measure the amount of fluid consumed (FLUID).
- $\text{SWEAT RATE FOR 1 HOUR} = \text{PREWEIGHT} - \text{POSTWEIGHT} + \text{FLUID}$

emergency response personnel should be educated about the possibility of EAH.

COLD INJURY

Individuals participating in winter sports performed for a long periods at high levels of exertion are at an increased risk for cold injury, primarily hypothermia and frostbite.⁷¹⁻⁷⁴

HYPOTHERMIA**Pathophysiology**

Hypothermia, defined as a core body temperature of less than 35°C (95°F), is the most common cause of weather-related death, with individuals 65 years and older being at greatest risk.⁷⁵⁻⁷⁹ Mortality increases proportionately with the severity of the hypothermia.⁸⁰ Dropping the core temperature results in nerve conduction delay, which leads to mental status changes, global depression of function, ataxia, and muscle stiffness. As hypothermia is neuroprotective, full neurologic recovery can occur even at remarkably low core temperatures.⁸¹⁻⁸⁴ Hypothermia results in electrocardiogram (ECG) changes, including abnormal repolarization, atrial fibrillation, and the J wave (a positive deflection at the QRS and ST-segment junction).^{85,86} With dropping core temperature, there is sequential prolongation of the PR, QRS, and QT intervals.⁷⁹ Vasoconstriction causes a relative hypervolemia, and leads to the “cold diuresis” that may in turn result in hypovolemia.

Heat Balance

To avoid functional impairment in cold environments, core temperature must be maintained within a 4°C range by increasing heat production and decreasing heat loss.

Body heat production, or thermogenesis, is achieved in four major ways.⁸⁷ The resting (or basal) metabolic rate refers to energy expenditure at rest in a thermoneutral environment. During exercise 75% of muscle metabolic energy is released as heat, and 25% is converted to work. The absorption, breakdown, and storage of ingested food produce heat, called the “thermic effect,” and provides a minimal proportion of total body heat production.⁸⁷ “Thermoregulatory thermogenesis” refers to shivering that can increase the metabolic rate up to five times the resting state.⁸⁷⁻⁸⁹ The amount of shivering is proportional to the amount of cold stress experienced, but can be maintained only

as long as glycogen stores are available.⁸⁸ Shivering ceases at a core temperature of 30°C.^{82,84}

About 90% of heat loss is through the skin, which is controlled by vasoconstriction. About 10% is lost through the lungs, which varies depending on the type and intensity of activity. Heat loss increases when an athlete is wet.⁸⁵ Radiation accounts for about 60% of total heat loss.⁷⁵ Radiation losses in children are higher due to their greater surface area-to-mass ratio.⁹⁰

Other Factors Contributing to Thermal Balance

Certain medical conditions such as hypothyroidism, hypoadrenalism, hypopituitarism, and diabetes can impair the body’s ability to produce heat.^{75,79,88} Altered heat balance occurs in the elderly, and in individuals with multiple sclerosis, Parkinson disease, certain skin conditions, and those taking certain drugs.^{75,91} Ethanol, the substance most commonly associated with hypothermia, leads to heat loss through peripheral vasodilation.^{75,91-93} Phenothiazine drugs impair shivering thermogenesis.⁹⁴ Medications, such as benzodiazepines, barbiturates, and tricyclic antidepressants, decrease centrally mediated thermoregulation.⁹⁵

Diagnosis

The diagnosis of hypothermia requires an accurate and reliable measurement of the core temperature. An esophageal probe placed in the lower one-third of the esophagus is preferred if there is a secure airway. If an esophageal probe is not available or if the airway is not secured, an epitympanic probe is recommended, although these can give falsely low measurements in the field. Rectal or bladder thermometers should not be used until the patient is in a warm environment.^{82,84} Oral or axillary probes do not reliably reflect core temperature.

In mild hypothermia (32°C to 35°C or 90°F to 95°F), the extremities are cool and pale, and performing fine movements of the hands is difficult. Shivering is evident. Neurologic exam reveals listlessness, confusion, disorientation, dysarthria, and ataxia. The person with moderate hypothermia (28°C to 32°C or 82°F to 90°F) exhibits markedly altered mental status, dysarthria, apathy, and amnesia. The mental status may deteriorate into stupor and coma. Deep tendon reflexes diminish and become absent. Vital signs become unstable; hypotension and bradycardia emerge. Shivering slows and ceases. Death will ensue without treatment.^{82,96} The person with severe hypothermia (<28°C or 82°F) is often comatose and may appear dead. The pupils may be fixed and dilated. Hypotension and bradycardia are severe; the respiratory rate decreases, and the vital signs may be difficult to obtain. Ventricular fibrillation and asystole are common. The muscles become rigid, and oliguria occurs. Persons with severe hypothermia have the highest mortality rate. Some investigators include a fourth category, profound hypothermia, which is defined as a core temperature less than 24°C or 20°C.⁸²

Laboratory Findings

ECG is helpful in diagnosing cardiac manifestations. Laboratory evaluation should include serum electrolytes (e.g., calcium, magnesium), amylase, and lipase, and blood glucose, arterial blood gas, a complete blood cell count, BUN, creatinine, prothrombin

time/international normalized ratio, partial thromboplastin time, and fibrinogen.⁹⁵

Treatment

Treatment depends on the severity of the hypothermia and the treatment setting. Rewarming is the primary treatment for hypothermia. Persons with mild hypothermia may be treated with passive rewarming, consisting of insulating the patient and allowing thermogenesis to increase the core temperature. All wet clothes should be immediately replaced with dry garments.^{82,84} Persons with moderate hypothermia require active external rewarming with care to avoid burns to the skin. Rewarming the trunk before the limbs is recommended to avoid *afterdrop*, in which core temperature drops precipitously because of shunting of cold blood away from the periphery to the core. Examples of methods for active external rewarming include blankets or electrical, chemical, or forced air heat packs.⁹⁷ A bath or a hot shower should not be used for rewarming as these can lead to cardiovascular collapse.^{82,84} The clothing of persons with a markedly altered mental status should not be removed; rather, warm blankets should be placed over them.⁸⁵ Care should be observed when moving patients with severe hypothermia to prevent the precipitation of ventricular fibrillation.

Patients with severe hypothermia require active internal rewarming using warmed IV fluids (38°C to 42°C) and warmed fluids instilled into the gastrointestinal tract. In severe hypothermia, extracorporeal rewarming appears to increase survival.^{83,97}

Extracorporeal rewarming refers specifically to shunting blood from the body, usually by cardiopulmonary bypass or venoarterial or venovenous shunt (extracorporeal membrane oxygenation [ECMO]), to a machine that rewarms the blood and returns it to the body. Extracorporeal rewarming is highly invasive and should be carried out only in centers with experience in these techniques. Patients with severe hypothermia can have profound bradycardia; carotid pulses should be checked for 60 seconds and cardiopulmonary resuscitation should begin if the pulse is absent.^{82,84} Intensive care monitoring is essential in cases of moderate to severe hypothermia.

Prevention

Athletes should know the weather conditions and dress appropriately. Inner layers ideally are composed of materials that wick moisture from the skin. Outer layers should allow moisture to transfer to the air, where it can evaporate without significant heat loss.⁸⁸ Head covering can diminish heat loss up to 50%.⁹⁸ Thick, loose socks are essential; socks should be changed frequently during prolonged exposure so the feet stay dry.⁸⁸ Athletes undertaking long training routes in remote areas should both train with a partner and carry a cell phone. Athletes participating in winter sports must pay close attention to energy stores; increased intake of carbohydrates is recommended. Adequate hydration is important to avoid hypovolemia from cold diuresis. Athletes participating in winter sports should be aware of the warning signs of hypothermia, and if they find themselves lost in the cold, they should remember that their chances for survival are better if they take temporary shelter and conserve heat than if they continue to trek in the cold.

FROSTBITE

Pathophysiology

The military and mountaineering have traditionally been the groups with higher risk of frostbite.⁹⁹ In frostbite, tissues of an extremity or appendage (nose, ears, penis) become frozen.⁷⁹ Frostbite progresses through four pathologic phases.^{100,101} In the prefreeze phase, vasoconstriction occurs as heat is lost in an effort to shunt blood to the core. Vasodilation will intermittently relieve this vasoconstriction approximately every 5 to 7 minutes in a phenomenon known as *cold-induced vasodilation*.^{102,103} At the freeze–thaw phase, ice crystals form intracellularly, altering the protein structure and causing cell dehydration and shrinkage. Inflammatory mediators are released and accumulate. Vascular stasis occurs and endothelial damage occurs, causing blood leakage and thrombus formation. Affected tissues undergo progressive ischemia leading to cell death.^{100,101} With continued exposure to the cold environment, frostbite progresses from distal to proximal and from superficial to deep.¹⁰⁴

Diagnosis

Frostbite patients present with a cold and sometimes frozen extremity with numbness and paresthesias. Frostbite has been classified into four degrees of injury.^{101,105,106} In first-degree injury, the affected area becomes firm and yellow to white, with significant edema. Second-degree frostbite will additionally exhibit characteristic clear to milky-appearing blisters. In third-degree injury, smaller hemorrhagic blisters indicate damage to the deeper dermis. Fourth-degree frostbite includes damage to the bone and muscle with gangrene and mummification likely.

Laboratory

Technetium (Tc) bone scanning within the first 5 days following rewarming reliably predicts which persons with frostbite would eventually require amputation and the extent of that amputation.^{107,108}

Treatment

Treatment of hypothermia should be the first priority.^{79,100,101} The mainstay of treatment of frostbite is rewarming, which will lead to significant pain, marked edema, and hyperemia. Rewarming should not be initiated in the field if refreezing is a possibility, because refreezing can cause more extensive tissue injury. If conditions in the field and during transport are such that the risk of refreezing is minimal, then the thawing process should be undertaken.¹⁰¹ Rapid rewarming is performed in warm water with a temperature in the range of 37°C to 42°C (98.6°F to 107.6°F) with added antiseptic solution.^{101,105} Care must be taken to ensure that the water temperature remains constant, which can be facilitated by keeping the water continually circulating. Rewarming should proceed until the extremity becomes red–purple and the skin is pliable, which typically occurs in 30 to 45 minutes. The affected tissue is often quite fragile; care must be taken to avoid having the extremity touch the sides of the water vessel to prevent tissue damage.¹⁰¹ The injured extremity should be gently cleaned and allowed to air dry. Dry bulky dressings help to minimize any further tissue trauma, and the extremity

should be elevated. Optimal management of blisters is controversial.^{101,105,106} Blisters should not be débrided in the field. Once in the hospital setting, needle aspiration may be performed on any clear blisters but not on hemorrhagic blisters. Topical aloe vera, a thromboxane inhibitor, in cream or gel form, is a recommended adjunctive treatment.^{101,105}

The rewarming process is extremely painful, and opioid analgesics are required for pain control.^{101,105} Ibuprofen, 12 mg/kg per day in two divided doses to a maximum of 2400 mg/day, should be given to inhibit inflammation.¹⁰¹ Tetanus prophylaxis should be administered per standard guidelines. Prophylactic antibiotics are not recommended. Antibiotics should only be given if signs or symptoms of infection/sepsis develop.¹⁰¹

Because part of the pathophysiology of frostbite involves microthrombi formation, thrombolytic therapy can be instituted in severe cases of frostbite. The use of an IV or intra-arterial tissue plasminogen activator within 24 hours of thawing has been shown to be of potential benefit in salvaging tissue at risk for necrosis.^{109,110} However, treatment with a tissue plasminogen activator has significant potential risks, and should only be undertaken at a medical facility with intensive care monitoring and experience in using it in cases of frostbite. Vasodilators have been used for the treatment of frostbite. The use of IV iloprost is recommended, but this medication is currently not available in the United States.^{105,109}

Surgical intervention with amputation should not be undertaken too early in the course of frostbite. Necrotic tissue demarcation can take up to 3 months. Most experts recommend early amputation only in cases of sepsis due to frostbitten tissue.^{101,107,108} Pain, burning, and electric shock-like sensations are the result of an ischemic neuritis and can persist for weeks to months. Long-term sensory deficits occur in virtually all persons with frostbite.¹⁰⁰

Prevention

Measures to prevent frostbite are the same as for preventing hypothermia.^{79,100,101,104,105} In addition, in extremely cold weather, mittens should be worn instead of gloves. People exercising in the cold should know the signs and symptoms of frostbite and check for them frequently. They should move to a sheltered environment at the earliest signs of frostbite.

HIGH-ALTITUDE ILLNESS

High-altitude illness (HAI) usually occurs at levels of 2500 m or greater, although susceptible individuals may develop symptoms as low as 2000 m. At high altitudes, athletes are subject to lower oxygen tension and air resistance, greater exposure to ultraviolet radiation, drier air, and colder temperatures. The rate of ascent is more strongly associated with the risk of HAI than absolute altitude. Although no precise definition of “rapid ascent” exists, an ascent to greater than 2400 m in the first day, or greater than 600 m/day above 2400 m, would be rapid enough to induce HAI.¹¹¹

The use of hypoxia or altitude as a training modification to improve athletic performance is beyond the scope of this chapter, but has been covered in recent reviews.¹¹²⁻¹¹⁴

Physiologic Adaptations to Altitude

At altitude, hypoxia induces cerebral arterial dilatation and venous occlusion, resulting in increased cerebral perfusion pressure, which stimulates hyperventilation, resulting in respiratory alkalosis.¹¹⁵⁻¹¹⁹ Pulmonary vasoconstriction increases pulmonary perfusion pressure. The affinity of hemoglobin for oxygen decreases, favoring increased oxygen delivery to the tissues. Within 48 hours the kidneys increase bicarbonate excretion (in response to the alkalosis) and increase erythropoietin production.¹¹⁵

HIGH-ALTITUDE HEADACHE, ACUTE MOUNTAIN SICKNESS, HIGH-ALTITUDE CEREBRAL EDEMA

Pathophysiology

High-altitude headache (HAH) is a common isolated finding in unacclimatized individuals ascending to altitude, and is likely due to mild cerebral edema.^{120,121} Increased pulmonary vascular pressure may trigger a mismatch in ventilation to perfusion, resulting in the mild dyspnea, whereas both dyspnea and hypoxia likely disrupt normal sleep.^{122,123}

Diagnosis

Acute mountain sickness (AMS) is defined as HAH accompanied by at least one other symptom: nausea, vomiting, shortness of breath, lethargy, sleep disturbance, and anorexia.^{121,124} The Lake Louise Consensus Scoring System has been widely used in the clinical diagnosis of AMS in adults (Box 21.5).¹²⁵⁻¹²⁷ A Children’s Lake Louise Score has been developed for young children.¹²⁸ A Lake Louise score of 3 or greater is consistent with a diagnosis of AMS. High-altitude cerebral edema (HACE), a potentially fatal global encephalopathy, is the end stage of AMS. HACE is diagnosed clinically in persons who ascend rapidly to a high altitude and consequently experience headache, mental status changes (e.g., confusion, difficulty concentrating, stupor, and drowsiness), and ataxia. Clinicians should be vigilant regarding persons with apparent AMS who continue to worsen, because deterioration from mild AMS to HACE may be swift. Patients with HACE typically present a nonfocal neurologic exam, although hemiparesis, hemiplegia, papilledema, retinal hemorrhage, and cranial nerve deficits have been reported.¹²⁹⁻¹³¹ A suggested simple test for subtle ataxia is to have the patient heel-toe walking in a straight line followed by a 90-degree turn; look for any unsteadiness.¹³¹

Risk factors for AMS include a history of AMS, a history of migraine, the use of alcohol, increased body mass index, elevated blood pressure, psychological stress, and inexperience with high altitudes.¹³²⁻¹³⁴ Greater physical fitness is not protective, and vigorous activity increases the risk for AMS.¹³⁵⁻¹³⁸ Lower arterial oxygen saturation with exercise at low altitude, and arterial desaturation in response to hypoxic exercise (hypoxic ventilatory response [HVR]), appear to be markers for susceptibility to AMS.^{132,137,139-141}

Laboratory Tests

Patients with altered mental status or ataxia should have magnetic resonance imaging (MRI) of the brain, but treatment of HACE

BOX 21.5 Lake Louise Consensus Scoring System for Acute Mountain Sickness

Self-Reported Symptoms

Headache

- 0 = None
- 1 = Mild
- 2 = Moderate
- 3 = Severe, incapacitating

Gastrointestinal Symptoms

- 0 = Good appetite
- 1 = Poor appetite or nausea
- 2 = Moderate nausea and/or vomiting
- 3 = Severe nausea and/or vomiting

Fatigue and/or Weakness

- 0 = None
- 1 = Mild
- 2 = Moderate
- 3 = Severe

Dizziness or Light-Headedness

- 0 = None
- 1 = Mild
- 2 = Moderate
- 3 = Severe, incapacitating

Difficulty Sleeping

- 0 = Slept as well as usual
- 1 = Did not sleep as well as usual
- 2 = Woke many times; poor night's sleep
- 3 = Could not sleep at all

Clinical Findings

Changes in Mental Status

- 0 = No change
- 1 = Lethargy/lassitude
- 2 = Disorientation/confusion
- 3 = Stupor/semiconsciousness
- 4 = Coma

Ataxia

- 0 = None
- 1 = Use of balancing movements
- 2 = Steps off the line
- 3 = Falls down
- 4 = Unable to stand

TOTAL SCORE

See references 115, 127.

should not be delayed waiting for imaging. Brain MRI reveals vasogenic cerebral edema that may involve both gray and white matter, especially the splenium of the corpus callosum, and microhemorrhages.^{119,142,143}

Differential Diagnosis

HAH is easily distinguished from migraine by its rapid resolution with supplemental oxygen.¹³⁰ Ailments that may resemble AMS include viral illness, a hangover, dehydration, exhaustion, hypothermia, or adverse effects of medication. Other conditions

that may mimic HACE include hypoglycemia, stroke, transient ischemic attack, hypothermia, central nervous system infection, carbon monoxide poisoning, drug or alcohol use, complex migraine, and diabetic ketoacidosis.

Treatment

Most cases of HAH and AMS can be treated symptomatically with supportive measures including rest, oxygen, avoidance of exertion, attention to hydration, and analgesics for headache. Treatment guidelines for AMS are detailed in Table 21.2.^{116,117,120,137,139,144,145} The most effective treatment for all types of HAI, including AMS, is immediate descent to lower altitudes. The urgency for descent will depend on the severity of illness. Individuals with HACE and high-altitude pulmonary edema (HAPE) should descend immediately.

Acetazolamide is the mainstay of pharmacologic treatment of AMS for both adults and children. Acetazolamide is contraindicated for individuals allergic to sulfonamides. Dexamethasone, alone or in combination with acetazolamide, is also effective in treating moderate AMS and HACE. Currently, there are insufficient data to recommend other medications, including antioxidants, magnesium, ginkgo biloba, sildenafil, and spironolactone, for the treatment of HAI.¹⁴⁶⁻¹⁴⁹ Temazepam, zolpidem, or zaleplon may be used as an adjunct to therapy in persons with AMS and significant sleep disturbance.¹⁵⁰

Return to Play

It is essential that the athlete be entirely symptom-free before resuming training or play at altitude.^{117,145} Regimens for graded ascent are suggested in recent reviews.^{111,144,151} Medication prophylaxis for susceptible persons should include acetazolamide, 125 to 250 mg twice a day, beginning 24 hours prior to ascent, and continuing for 2 to 4 days upon arrival at high altitude. Dexamethasone, 2 mg every 6 hours or 4 mg every 12 hours, can be used alone or in combination with acetazolamide and continued for the duration of the ascent to avoid rebound symptoms.¹¹⁷ Persons with permanent neurologic sequelae following HACE should never return to a high altitude.

HIGH-ALTITUDE PULMONARY EDEMA

Pathophysiology

HAPE presents within 2 to 5 days after arrival at a high altitude with dyspnea, exercise intolerance, and a persistent dry cough.¹⁵² Tissue changes with HAPE include patchy pulmonary vasoconstriction that results in elevated capillary pressure, damage to pulmonary capillaries with resultant leakage, which may induce inflammation, and hypoxia as a result of ventilation/perfusion mismatch.^{115,116,153}

Diagnosis

Production of pink, frothy sputum, although classically described, is not common.¹⁵⁴ Auscultation reveals rales. Peripheral cyanosis and edema may be present in severe cases. Fifty percent of persons with HAPE also have AMS, and up to 14% have HACE.¹¹⁵ HAPE accounts for the majority of deaths from high-altitude illness.¹¹⁵

TABLE 21.2 Treatment of High-Altitude Illness

Intervention	AMS—mild (LLS < 5)	AMS—moderate to severe (LLS ≥5)	HACE	HAPE
Supportive	Rest, hydration, analgesics	Rest, hydration, analgesics	Rest, hydration, analgesics	Rest, hydration, analgesics
Descent	Usually not necessary; consider descent 500–1000 m if symptoms fail to improve after 24 h of treatment	Immediately descent 500–1000 m	Descend as soon as possible to lowest possible altitude at which symptoms improve	Descend as soon as possible to lowest possible altitude at which symptoms improve
Oxygen	May be administered to alleviate symptoms	Low-flow O ₂ (1–2 L/min) or utilize portable hyperbaric chamber if descent not possible	2–4 L/min to maintain O ₂ sat ^a >90%	2–6 L/min to maintain O ₂ sat >90%
Medications	Acetazolamide 125–250 mg PO every 12 h Anti-emetics if necessary	Acetazolamide 250 mg every 12 h AND Dexamethasone 4 mg by PO, IV, or IM every 6 h	Dexamethasone, load with 8 mg PO, IV, or IM, then 4 mg every 6 h If descent is delayed, add Acetazolamide 250 mg every 12 h	Nifedipine 10 mg by mouth initial dose, then 30 mg extended-release every 12 h Consider Salmeterol inhaled 125 µg every 12 h Consider dexamethasone if symptoms suggest HACE

See references 116, 117, 120, 137, 139, 144, 145.

^aArterial oxygen saturation.

AMS, Acute mountain sickness; HACE, high-altitude cerebral edema; HAPE, high-altitude pulmonary edema; IM, intramuscular; IV, intravenous; LLS, Lake Louis Score; O₂, oxygen; PO, per os (by mouth).

Laboratory Findings

Arterial oxygen (O₂) saturation less than 90% is characteristic of HAPE. Chest x-ray reveals engorgement of the pulmonary arteries and patchy infiltrates, although diffuse lung involvement may be seen in more severe cases. Arterial blood gas analysis reveals severe hypoxemia (30 to 40 mm Hg) and respiratory alkalosis.¹⁵² ECG may reveal right ventricular strain (e.g., right-axis deviation and p-wave abnormalities) and a right bundle branch block.¹¹⁵

Differential Diagnosis

Other causes of respiratory distress that may resemble HAPE include asthma, bronchitis, pneumonia, heart failure, myocardial infarction, and pulmonary embolus.

Treatment

Treatment guidelines for HAPE are summarized in Table 21.2. Patients whose O₂ saturation either fails to improve to 90% or greater who have evidence of HACE should descend immediately. Adding nifedipine, 30 mg by mouth twice a day, may be useful when either descent is not possible or oxygen is not available.^{152,155} The administration of high-dose salmeterol, five puffs (125 µg) twice a day may improve oxygenation.¹⁵⁶

Return to Play

Guidelines for RTP after AMS are applicable to persons with HAPE. Prophylaxis with dexamethasone, 8 mg twice a day, should commence 24 hours prior to ascent.¹⁵⁷ Tadalafil, 10 mg by mouth twice a day, begun 24 hours prior to ascent, is almost as effective as dexamethasone in preventing HAPE.¹⁵⁸ In a randomized controlled trial, nifedipine 30 mg twice a day afforded greater

reduction in HAPE recurrence than inhaled salmeterol 125 µg twice a day.¹⁵⁶

Advice for the Team Physician Traveling to Altitude

The physician accompanying a team to altitude for training or competition can implement measures to both minimize performance decrements and the risk of HAI. The team physicians and athletic training staff should review symptoms of HAI before traveling and regularly remind players to report symptoms, particularly headache, immediately.¹⁵⁹

The strongest predictor of performance decrement at altitude is the HVR, a measure of how well the individual's respiratory apparatus compensates for lower oxygen tension. Measuring the HVR could identify athletes who may require individualized acclimatization, but doing so requires special equipment, which is not always readily available.

Athletes with positive risk factors for HAI, including a prior history of HAI, a history of migraine, a history of anemia or low iron stores, an increased BMI, and elevated blood pressure, should receive prophylaxis. Prophylaxis and treatment of HAI must be chosen with consideration of applicable antidoping regulations. Both dexamethasone and acetazolamide are on the World Anti-Doping Agency (WADA) banned drug list.¹⁶⁰ An acceptable prophylactic regimen is ibuprofen 600 mg by mouth 3 times/day begun 1 to 2 days before ascent for AMS.¹⁶¹ The authors recommend the coadministration of omeprazole 20 mg by mouth once daily to avoid gastritis. Prophylaxis of HAPE may consist of nifedipine or tadalafil. WADA allows a maximum daily dose of salmeterol of 200 µg, which also may be used for HAPE prophylaxis, equivalent to four puffs twice a day (which is 20% less than the

recommended dose for prophylaxis).^{144,160} The team physician should always contemporaneously consult the latest applicable banned drug list.

The ideal approach to guard against the effects of altitude would be to acclimatize. The use of techniques, such as intermittent hypoxemic exposure (IHE), intermittent hypoxemic training (IHT), and live high, train low (LHTL), have been shown to improve the HVR, but require hypobaric or hypoxemia chambers that are not readily available to most teams.¹⁶² Recommended graded ascent acclimatization to altitudes above 3000 m consists of increasing sleep altitude by no more than 500 m to 600 m per day, and a day of rest for each additional 1000 m if possible.¹⁶³

Augmenting carbohydrates, such as pushing a carbohydrate-containing sport drink instead of water, may help to prevent performance decrements and dehydration.¹⁶⁴⁻¹⁶⁷ Individuals with a history of anemia or low iron stores should have their iron status rechecked several weeks prior to travel to altitude and should begin supplemental iron accordingly. Reducing the intensity of workouts and increasing recovery periods in the first 1 to 2 days at altitude may allow partial acclimatization.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Casa DJ, DeMartini JK, Bergeron MF, et al. National Athletic Trainer's Association position statement: exertional heat illness. *J Athl Train*. 2015;59(9):986–1000.

Level of Evidence:

III

Summary:

Evidence-based, best-practice recommendations for the prevention, recognition, and treatment of exertional heat illnesses (EHIs) and description of the relevant physiology of thermoregulation.

Citation:

Bergeron MF, Devore C, Rice SG. Council on Sports Medicine & Fitness and Council on School Health. Policy statement—climatic heat stress and exercising children and adolescents. *Pediatrics*. 2011;128(3):e741–e747.

Level of Evidence:

III

Summary:

Evidence-based summary of differences in response to exercise heat stress between child and adult athletes, with specific recommendations for avoiding heat illness for child athletes.

Citation:

Paal P, Gordon L, Strapazzon G, et al. Accidental hypothermia—an update: the content of this review is endorsed by the

International Commission for Mountain Emergency Medicine (ICAR MEDCOM). *Scan J Traum Rescuc Emerg Med*. 2016;24(1):111.

Level of Evidence:

III

Summary:

This paper provides an up-to-date review of the management and outcome of accidental hypothermia patients with and without cardiac arrest.

Citation:

McIntosh SE, Dow J, Hackett PH, et al. Wilderness Medical Society. Wilderness Medical Society practice guidelines for the prevention and treatment of frostbite: 2014 update. *Wilderness Environ Med*. 2014;25(suppl 4):S43–S54.

Level of Evidence:

III

Summary:

Evidence-based guidelines from an expert panel convened by The Wilderness Medical Society on the prevention and treatment of frostbite. The paper reviews pertinent pathophysiology, and discusses primary and secondary prevention measures and therapeutic management.

Citation:

Luks AM, McIntosh SE, Grissom CK, et al. Wilderness Medical Society practice guidelines for the prevention and treatment of acute altitude illness: 2014 update. *Wilderness Environ Med*. 2014;25:S4–S14.

Level of Evidence:

III

Summary:

Best practices guidelines from an expert panel convened by the Wilderness Medical Society for the prevention and treatment of acute mountain sickness, high-altitude cerebral edema, and high-altitude pulmonary edema. These guidelines present the main prophylactic and therapeutic modalities for each disorder and provide recommendations about their role in disease management. Recommendations are graded based on the quality of supporting evidence.

Citation:

Bärtsch P, Swenson ER. Clinical practice: acute high-altitude illnesses. *N Engl J Med*. 2013;368:2294–2302.

Level of Evidence:

III

Summary:

An evidence-based clinical guideline on high-altitude illness, including a review of formal guidelines, as well as authors' expert recommendations.

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High Altitude Illness

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